



COMPUTATIONAL NEUROSCIENCE MODEL

# SynapseLab

## Synaptic Plasticity as Short-Term Memory

Explore how changing synaptic connection strengths temporarily retain information and trigger memory interference.



# Problem Statement



## CONVENTIONAL LIMITATION

### Static Explanations

#### Opaque Mechanisms

Memory dynamics remain abstract in standard textbooks.

#### Hidden Temporal Shifts

Learners cannot observe weight adaptation over time.



## THE INNOVATION

### SynapseLab Approach

#### Interactive Experimentation

Direct manipulation replaces static schematic diagrams.

#### Immediate Feedback Loop

Tuned learning inputs yield instant visual memory shifts.



# Technical Claim

“Changing synaptic connection strengths can temporarily store information, but conflicting information can modify those connections and cause memory interference.”



## Empirically Testable

Verifiable live trials directly validate synaptic updates.



## Interference Dynamics

Demonstrates how incoming associations overwrite prior weights.



## Pedagogical Scope

Clear educational simplification, not a biological simulation.



# What is Synaptic Plasticity?

## Connection Strength Modulation

Synaptic efficacy shifts between interacting neurons during transmission.

## Experience-Driven Adaptation

Modifications occur dynamically through repeated activity and exposure.

## Short-Term Memory States

Transient plasticity provides a natural substrate for temporary memory buffers.

## SynapseLab Representation

Memory traces represented directly by continuous, mutable weight values.

# What is synaptic plasticity?

Synaptic plasticity describes changes in the strength of connections between neurons. In computational models, changing connection values can provide a mechanism for representing information over time.

01

### Association

Suppose a system has learned that a particular input is associated with one outcome. The connection representing that association becomes relatively strong.

02

### Conflicting evidence

Now introduce a second, conflicting association. In our simplified model, the new evidence strengthens one connection while reducing the relative strength of the previous one.

03

### Interference

As conflicting examples accumulate, the original association may lose dominance. This gives us a measurable way to observe memory interference.

MODEL MECHANISM

### What changes during learning?

TOY MODEL

$$w_A \leftarrow w_A + \eta$$

Here,  $w$  represents connection strength and  $\eta$  represents the learning rate. The actual SynapseLab implementation uses a simplified bounded update rule so the learner can directly observe the effect.

BDH CONNECTION

### From Attention to Synaptic Memory

This module gives an educational view of how a brain-inspired architecture can treat changing internal state as a form of memory while processing a sequence.

CONCEPTUAL MODEL

04



# Short-Term Memory Model

SYNAPTIC WEIGHT UPDATE RULE

Discrete Step Model

$$\text{New Memory} = \text{Old Memory} + (\eta \times \text{Learning Signal})$$

## Learning Rate ( $\eta$ )

Controls the scaling impact of incoming training evidence on stored weights.

## Velocity Dynamics

High  $\eta$  delivers rapid adaptation; low  $\eta$  protects established memories from decay.

## Educational Scope

A transparent conceptual sandbox isolating the core principle of plasticity.



# Interactive Experiment

## Baseline Weight Distribution

$\eta = 0.10$

CAT → MEOW (Initial Dominant)

0.80



CAT → BARK (Suppressed)

0.20



Normalized associative weights model competitive recall states.

### 1. Baseline Encoding

Reinforce CAT → MEOW to establish dominant synaptic weights.

### 2. Introduce Interference

Present conflicting CAT → BARK trials to perturb connection values.

### 3. Observe Redistribution

Track real-time reallocation of synaptic transmission strengths.



# Memory Interference



## Signal Competition

Incoming evidence for CAT → BARK directly competes against established CAT → MEOW synaptic pathways.



## Synaptic Overwrite

Cumulative contradictory cues systematically lower previous weights, modifying stored associative memory.



## Empirical Validation

Provides clear quantitative demonstration of retroactive interference in artificial neural synapses.



# Effect of Learning Rate

## Low Learning Rate ( $\eta$ )

**High Stability:** Synaptic connections shift incrementally, defending prior memory against noise.

Best suited for durable baseline retention across conflicting sessions.

## High Learning Rate ( $\eta$ )

**Rapid Adaptation:** Immediate synaptic updates overwrite prior states in fewer trials.

Elevates risk of catastrophic forgetting of established knowledge.

Interactive variable exploration: Learners adjust  $\eta$  sliders to inspect stability-plasticity trade-offs in real time.

[LIVE PARAMETER  
TUNING](#)



# BDH Connection

## CONCEPTUAL COMPUTATIONAL PIPELINE

1. Input Vector



2. Neural Activity



3. Synaptic Update



4. Temporary Memory



5. Sequence Recall

### State-Based Retention

Brain-inspired architectures employ dynamic internal states to store temporal sequence context.

### Plasticity Bridge

BDH integrates biological synaptic mechanisms directly with modern transformer representations.

### Pedagogical Scope

Educational conceptual illustration; not an official production BDH release.



# System Architecture

## Client Technology Stack



 Backend: Zero server footprint (100% in-browser computation).



### App.jsx

Central application state, parameter routing, and interactive hooks.



### NeuronGraph.jsx

Dynamic SVG network rendering synaptic connection strengths.



### BDHModule.jsx

Educational bridge linking synaptic plasticity to modern AI models.



# Live, Synthetic & Educational Components

## Live Computation

- Real-time memory updates
- Dynamic learning-rate scaling
- Competitive association shifts
- Instant graph recalculation

## Synthetic Data

- **CAT** → **MEOW**: Baseline cue
- **CAT** → **BARK**: Contradictory cue
- Minimalist toy pair dataset
- Controlled test environment

## Educational Visuals

- Network node topology
- Synaptic strength gauges
- Intuitive mental model design
- Transparent learning curve



# Limitations

 **Simplified Mathematical Formulation:** Memory update rule abstracts away biophysical multi-compartment dynamics.

 **Biological Complexity:** Does not capture ion-channel kinetics, vesicle depletion, or precise spike timing.

 **Synthetic Data Scale:** Relies exclusively on a compact 2-association toy dataset for demonstration.

 **Conceptual BDH Framing:** Serves as a high-level educational analogy rather than a production-grade benchmark.

 **Implementation Extent:** The demo does not reproduce the complete published BDH architecture.



# Research References

2022

**Kozachkov et al.**

*Robust and brain-like working memory through short-term synaptic plasticity*

2022

**Rodriguez, Guo & Moraitis**

*Short-Term Plasticity Neurons Learning to Learn and Forget*

2024

**Cognitive Neuroscience Consortium**

*The synaptic correlates of serial position effects in sequential working memory*

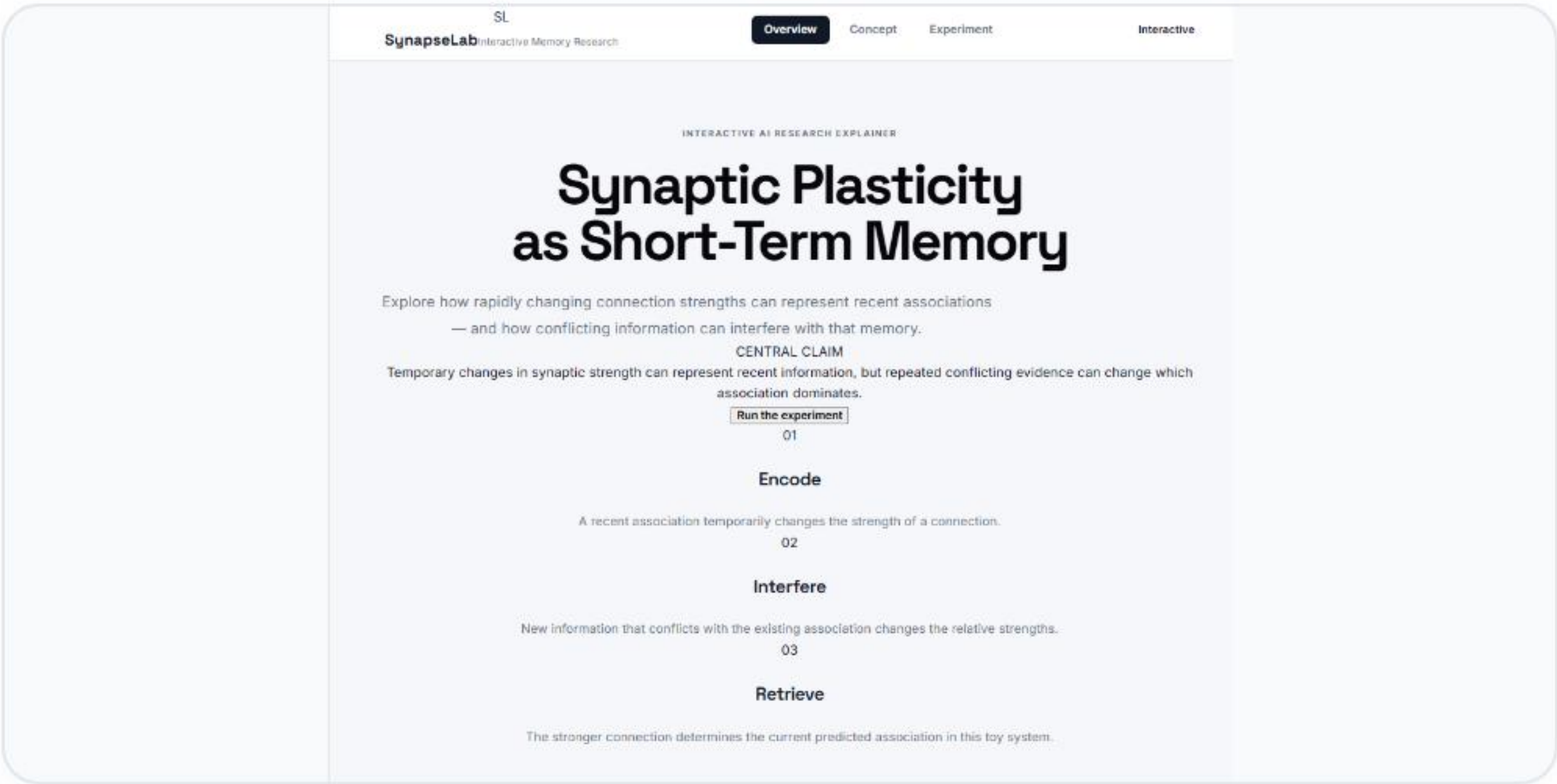
2025

**Kosowski et al.**

*The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*



# Demo & Conclusion



**LIVE PLATFORM**  
**SynapseLab Web Demo**

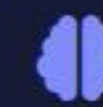
**OPEN SOURCE**  
**GitHub Repository**

**AI Disclosure:** AI assistance supported coding, documentation, and structural research synthesis.

## Key Takeaways

- Synaptic weights effectively serve as temporary memory repositories.
- Contradictory input patterns cause quantifiable memory interference.
- Learners directly explore parameter trade-offs in real time.
- Bridges core neurobiology with modern BDH architectures in an accessible form.



 DATAFORGE 2026

# Thank You!

SynapseLab: Synaptic Plasticity as Short-Term Memory

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Questions, feedback, and open discussion on neural dynamics and  
memory interference.



Interactive Web Demo



GitHub Repository



# Image Sources & Project Assets



**SynapseLab Interactive Explainer UI**  
image\_278a02.png • Overview, Central Claim & Encode/Interfere/Retrieve pipeline

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**SynapseLab Plasticity Mechanism Explainer**  
image\_279529.png • What is Synaptic Plasticity, Toy Model & BDH connection

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